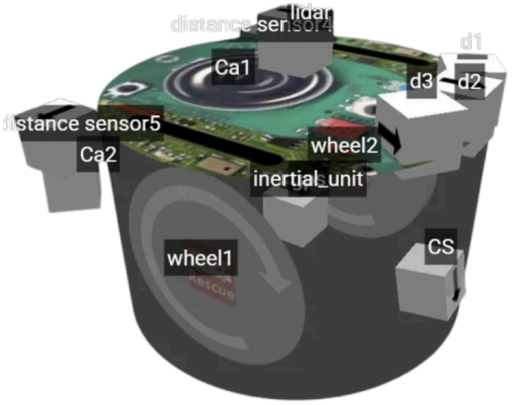
Rescue Simulation_Anyang_Additional Software Reference Documents

2. Robot Design

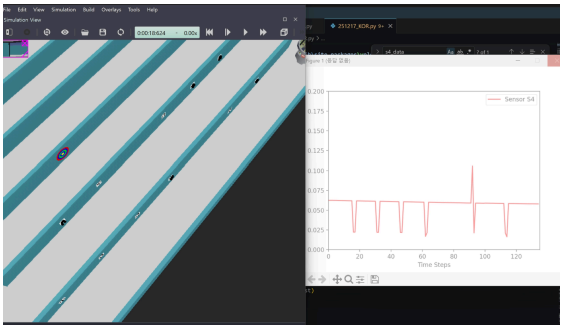
2-1. Robot configuration + sensors

Our robot is equipped with two cameras, a GPS, five distance sensors, one color sensor, an IMU (Inertial Unit), and a LiDAR.



One of the most distinctive features of our robot is that the cameras are mounted diagonally on the rear. Initially, we placed the cameras near the wheels, but they were too close to the victims to recognize them properly. To fix this, we tried moving them further inward, but they either interfered with the wheels or had their view blocked by the robot's body. Ultimately, by positioning the cameras on the rear diagonals, we secured a wider field of view and allowed the robot to detect victims much faster than a front-facing setup, maximizing the efficiency of our victim detection.

We also faced a lot of trial and error in distinguishing between real and false victims. At first, we expected that the readings from distance sensors 4 and 5—which measure the perpendicular distance to the walls—would drop sharply when passing a false victim. However, physical testing showed very little difference in the sensor values between real and false victims, so we switched to using LiDAR. But another challenge arose: the LiDAR had to be mounted on top to detect walls for navigation, while the victims were located near the center of the walls, making them invisible to the LiDAR.

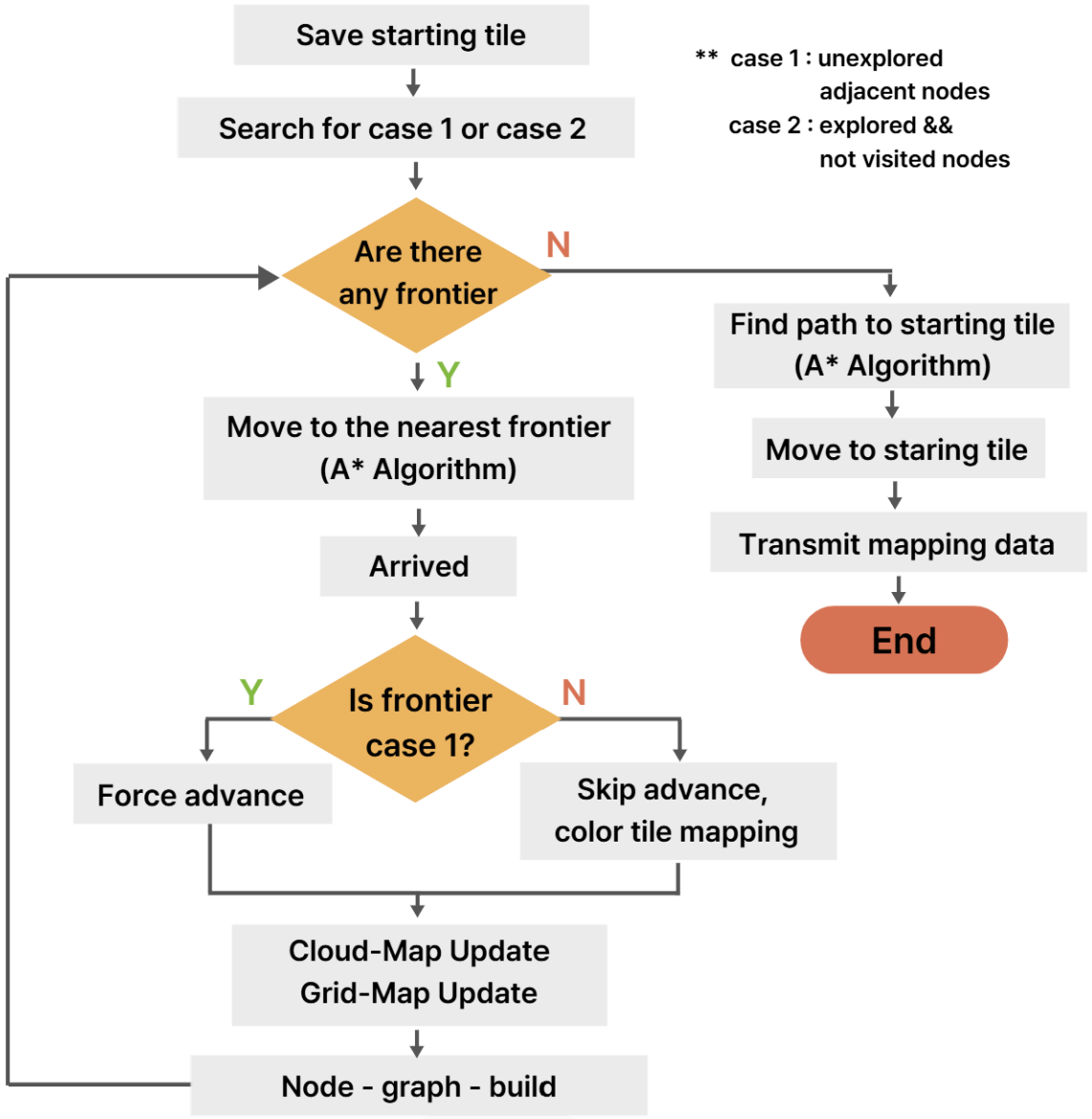


To overcome this, we decided to tilt the camera angles as much as possible. When the robot suspects a victim is present, it turns 15 degrees three times, capturing and analyzing the image at each angle. If the model determines that no victim is present in at least two out of the three checks, it classifies it as a false victim. This algorithm significantly improved our detection accuracy.

Our autonomous navigation and mapping system also relies on the organic integration of various sensors. First, the LiDAR detects surrounding walls to establish the structural framework of the map. The GPS tracks the robot's coordinates for mapping, utilizing both absolute and relative coordinates to construct node graphs for A^* navigation. Meanwhile, the IMU checks the robot's orientation and heading, playing a crucial role in aligning and matching real-time LiDAR data with the map. For navigation, distance sensors 1, 2, and 3 check for immediate obstacles and unmapped walls in real-time to avoid collisions. Lastly, the color sensor scans the floor to map and identify specific zones, such as swamps, holes, and colored tiles, helping the robot navigate safely and efficiently.

4. Navigation + Implementation

4-1. Overview



Our navigation program is built on **frontier-based exploration combined with the A* algorithm**. The overall flow begins with scanning for frontiers, selecting the nearest one, and navigating to it via the A* shortest-path algorithm. Once the robot reaches the frontier and explores the area, it repeats this cycle until no frontiers remain, at which point it returns to the starting tile and terminates.

Frontier selection relies on mapping data constructed from the GridMap and CloudMap. Each tile is subdivided into 9 nodes to form a graph, with edges connecting adjacent nodes. Two types of frontiers are defined: Case 1 frontiers are nodes adjacent to unexplored cells, while Case 2 frontiers are the center nodes of cells where `explored = True` but `visited = False`. To ensure Case 2

frontiers are visited preferentially, their distance weight is multiplied by a factor of 0.2.

Each Cell stores four pieces of information: wall, floor, `explored`, and `visited`. The `explored` flag indicates whether more than 90% of the CloudMap points for that Cell have been recorded, while `visited` indicates whether the robot has physically reached the center of that Cell.

Upon arriving at a frontier, the robot updates the full map and proceeds to select the next frontier. Once no frontiers remain, it returns to the starting tile and the run is terminated.

Several terrain-handling mechanisms are also in place. When the color sensor detects black, the robot identifies a hole, reverses, updates the floor data of the Cell ahead, and removes the corresponding nodes and edges from the graph. For tiles identified as swamp, the edge weights of all connected edges are multiplied by 20 to strongly discourage routing through those areas. Color tile zone classification is carried out when the robot reaches the center node of a tile. If the distance sensor detects a collision with an obstacle at the same target node three or more times, that node and its edges are deleted and the path is replanned. Finally, if the robot's navigation target remains unchanged for more than 50 ticks — indicating it may be stuck in a narrow passage between half-tiles that appears traversable in the graph but physically is not — the situation is classified as a stuck state, the relevant edge is removed, and the path is replanned.

4-2. Research and Analysis, Reliability Tests and quality assurance

At RoboCup Korea Open 2026, navigation was carried out using Depth-First Search (DFS) with each full tile treated as a single node. While this approach worked in straightforward grid layouts, it proved inadequate for paths composed of half-tiles or curved tiles, as the tile-granularity resolution made those geometries difficult to recognize and traverse. As a result, the algorithm could not be applied to Zone 4 at all, and the robot resorted to random driving in that zone without any structured navigation. Further issues included discontinuous LiDAR usage and difficulty returning to the starting tile within the time limit. Overall, the system was assessed as inefficient and in clear need of improvement.

To address these limitations, a new navigation architecture was designed around three core components: a graph structure subdividing each Cell into 9

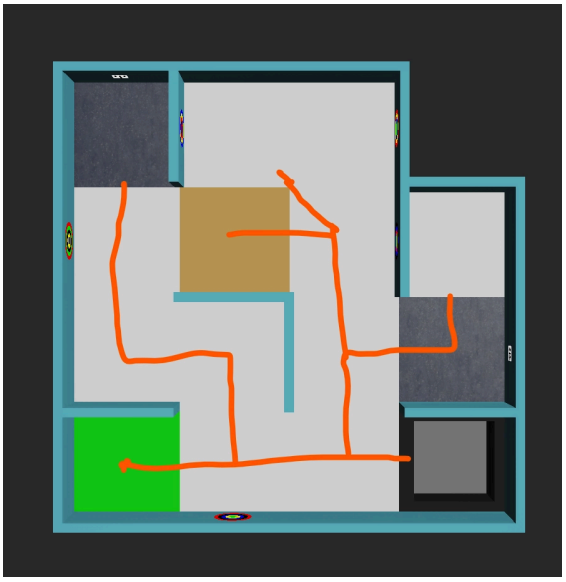
nodes, the A* shortest-path algorithm, and frontier-based exploration. A supporting mapping system was developed using a grid-based GridMap with LiDAR point cloud data from the CloudMap.

When generating nodes, the Cell values at each of the 9 node positions are checked. For edge and corner positions, all overlapping Cells are verified, and if any one of them is a wall, the node at that position is not created. Unexplored Cells (`explored = False`) contribute their adjacent nodes to the frontier list. The robot selects the nearest frontier and navigates to it, then advances 2 cm toward the unexplored Cell upon arrival to physically enter it, at which point `explored` is set to `True`.

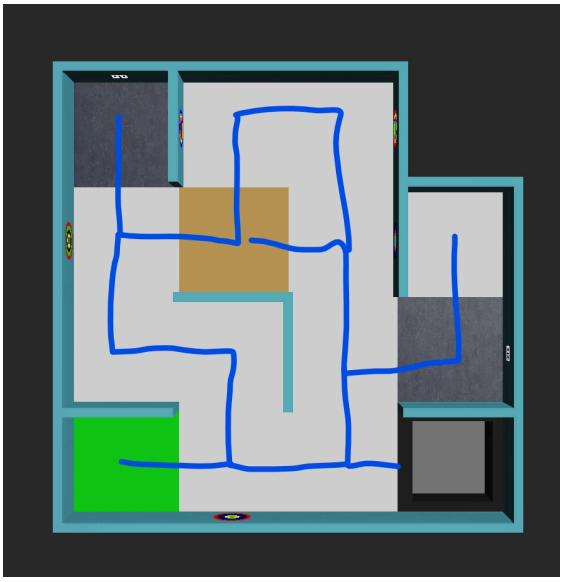
Issue 1: Premature `explored` Marking

During initial testing, it was observed that a Cell would be marked as `explored` even when the robot had only barely entered it, resulting in large portions of the map being left uncovered.

To resolve this, a `visited` flag was introduced. Unlike `explored`, `visited` is only set to `True` when the robot reaches the center of a Cell. The frontier list was accordingly updated to include the center nodes of Cells where `explored == True` and `visited == False`, ensuring the robot follows through to the Cell center rather than moving on after a shallow entry.



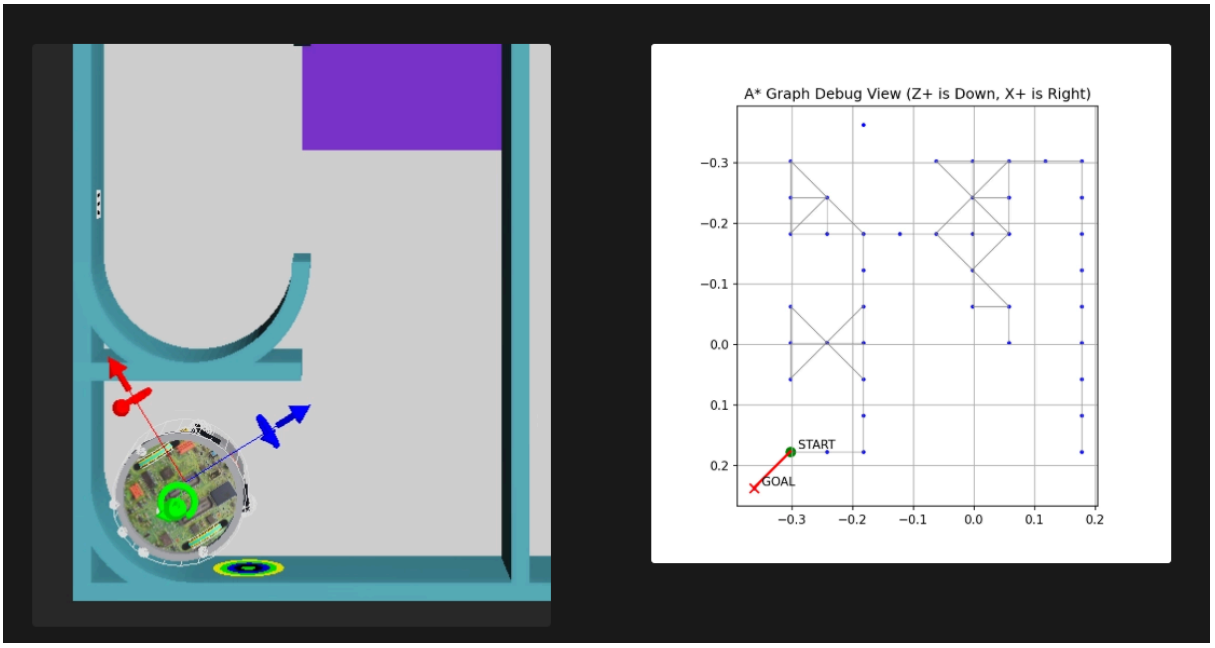
Map image showing robot traversal path before fix



Map image showing robot traversal path after fix

Issue 2: Wall Misdetection and Navigation to Curved Wall Vertices

Two related problems emerged in more complex environments. First, the wall mapping correction algorithm (described in a later section) occasionally caused flat walls to go undetected. Second, nodes were being generated at the vertex positions of curved walls, leading the robot to attempt navigation toward those geometrically inaccessible points.



World View and Node Graph Under Problematic Conditions

For flat walls, the obstacle-handling logic was applied directly: if the distance sensor flags the same target node three or more times, the node and its connected edges are deleted and the path is replanned. Curved walls, however, are harder to detect with the distance sensor, and GPS readings tend to fluctuate significantly when the robot slides along them. For these cases, a time-based fallback was implemented — if the same target node persists for more than 50 ticks, only the corresponding edge is deleted and the path is replanned, without removing the node itself. Although escape takes slightly longer in some edge cases, normal navigation is reliably recovered.

Issue 3: Vertex Looping When One of Four Adjacent Tiles Is Unexplored

A more subtle problem arose in situations where only one of four surrounding tiles remained unexplored. In such cases, the shared corner node at the center of the four tiles was typically selected as the frontier. If the robot failed to identify the direction of the unexplored tile, completing the 2 cm forward step upon arrival would not result in any tile being newly marked as `explored`, causing the same corner node to be repeatedly selected and the robot to

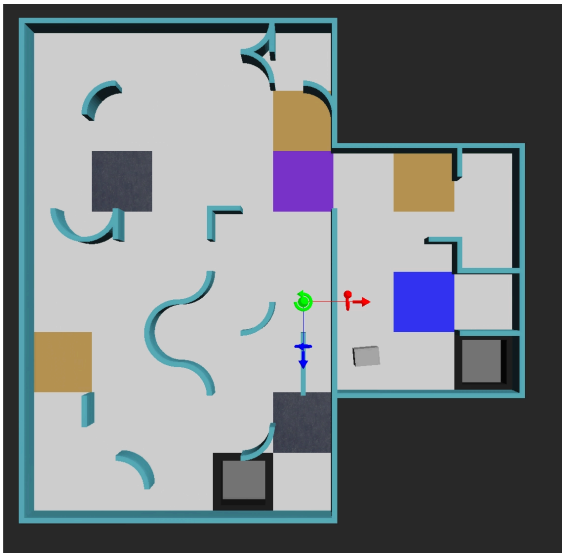
remain trapped in place. While the robot could occasionally escape if it happened to move in the right direction, the process was time-consuming and significantly reduced navigation efficiency.

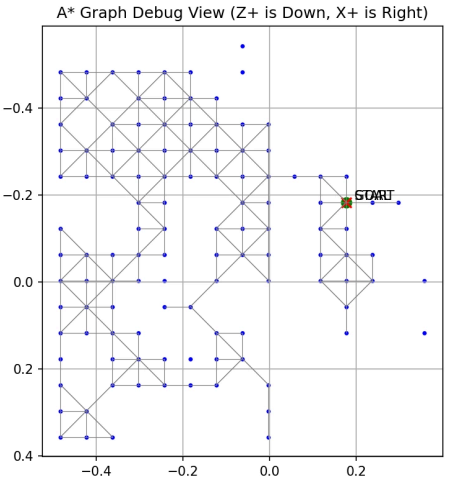
To address this, the method for setting `explored = True` was revised from a GPS-based approach to a CloudMap-based one. Under the previous method, `explored = True` and wall mapping were applied to the robot's current Cell and the four adjacent Cells based on the robot's GPS position. The revised method instead surveys a 5×5 area centered on the robot's current Cell and marks any Cell containing 10 or fewer `_UNDEFINED` cloud points in the CloudMap as `explored = True`, applying wall mapping accordingly. This allows the `explored` status to reflect actual sensor coverage rather than mere physical proximity.

In addition, map update timing was streamlined. Previously, updates were triggered from various points in the code, leading to redundancy and inconsistency. Updates are now limited to two events: upon arrival at a frontier, and upon obstacle detection. The frontier weighting scheme was also adjusted so that Case 2 frontiers (`explored && not visited`) are assigned a weight of 0.2× the distance, making them significantly more attractive than Case 1 frontiers (adjacent to unexplored areas) and prioritizing thorough Cell-center coverage over raw frontier expansion.

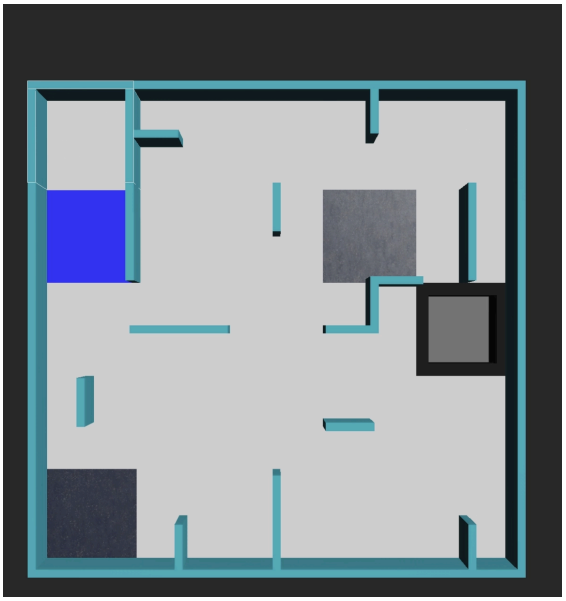
Following these changes, the vertex looping problem no longer occurs. The robot now navigates systematically toward Cell centers while reliably covering narrow and peripheral areas such as half-tile passages.

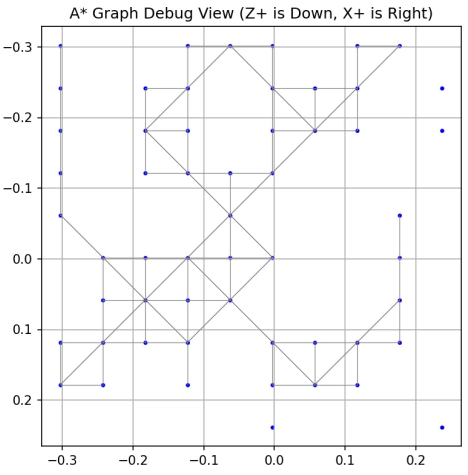
The self-designed map and its corresponding final node graph shown below confirm that the robot navigates successfully throughout the environment.





Map and node graph - Curved Wall Test





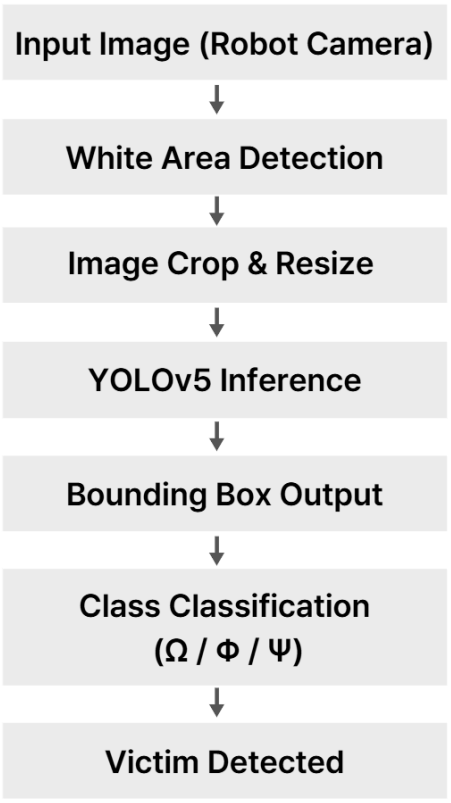
Map and node graph - Half-tile Wall Test

5. Victim Detection + Implementation

5-1. Architecture design (with diagrams such as flowchart, UML, pseudocode)

The victim detection system is built on the YOLOv5 architecture, a single-stage object detector based on a CNN backbone. The robot's camera first screens for white areas on walls as a pre-filter, since victim markers are always

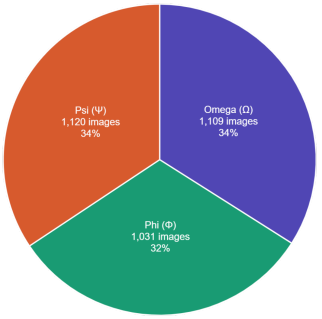
surrounded by white. If white is detected, the image is passed to the YOLOv5 model, which outputs bounding boxes and class labels for the three victim types: omega (Ω), phi (Φ), and psi (Ψ).



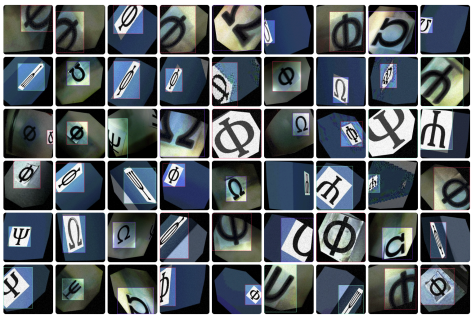
YOLOv5 Detection Pipeline Flowchart

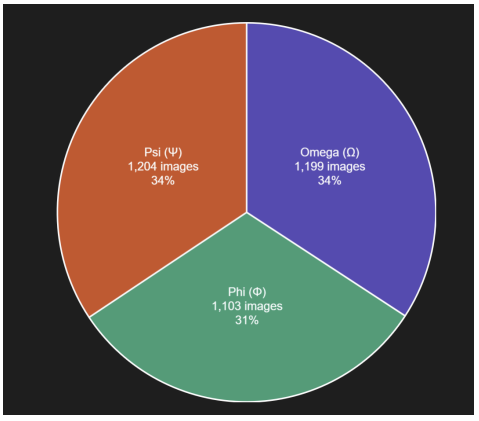
5-2. Research and Analysis

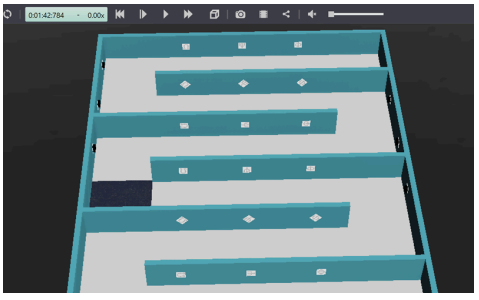
To identify victims, we used YOLOv5, a real-time object detection model based on a CNN architecture. YOLOv5 was selected for its fast inference speed and high accuracy in small-object detection, making it well-suited for the Webots environment where the robot must detect Greek letter symbols (Ω, Φ, Ψ) displayed on walls as victim markers. Training was conducted using the Ultralytics YOLOv5 framework on Google Colab with an NVIDIA A100 GPU, and data management



and preprocessing were handled through the Roboflow platform.
The dataset consisted of general Greek letter images sourced from Roboflow (omega: 674, phi: 567, psi: 586) and screenshots captured directly from the Webots map (omega: 435, phi: 464, psi: 534). For the screenshots, we varied the direction of sunlight to diversify the wall colors surrounding the victims, and captured images from different distances and angles to prepare the model for a wide range of situations. The final dataset was composed of omega: 1,109 images (34%), phi: 1,031 images (32%), and psi: 1,120 images (34%), keeping the class distribution balanced. All collected data was labeled, preprocessed, and augmented using Roboflow. Labeling was done manually by drawing bounding boxes around each victim and assigning the corresponding class. Preprocessing included Resize: Stretch to 640×640 and Auto-Adjust Contrast: Using Histogram Equalization. Augmentation settings were as follows: Outputs per training example: 3 / Flip: Horizontal, Vertical / Rotation: Between -45° and +45° / Shear: ±30° Horizontal, ±30° Vertical / Blur: Up to 6.5px / Noise: Up to 10% of pixels. After preprocessing and augmentation, the total number of images reached 7,731, split into train: 6,756 (approximately 70%), valid: 648 (approximately 20%), and test: 327 (approximately 10%).







To address these issues, two approaches were taken. First, unlabeled images were removed from the dataset. Previously,

images where the victim was barely visible or where the text appeared in colors other than black were kept in the dataset without labels, as they were not intended for training. However, we determined that these images were interfering with model training and real-world performance, so 158 unnecessary images were deleted. Second, we added data captured from the robot's actual point of view. While manually driving the robot in Webots, the camera automatically saved images whenever white areas were detected, since the area surrounding victim letters is white and serves as a reliable indicator of a victim's presence. Unlike the existing high-quality images, this newly collected data has lower resolution and blurrier text due to the nature of the robot's camera. After these deletions and additions, the dataset was updated to omega: 1,199 images (approximately 34%), phi: 1,103 images (approximately 32%), and psi: 1,204 images (approximately 34%).

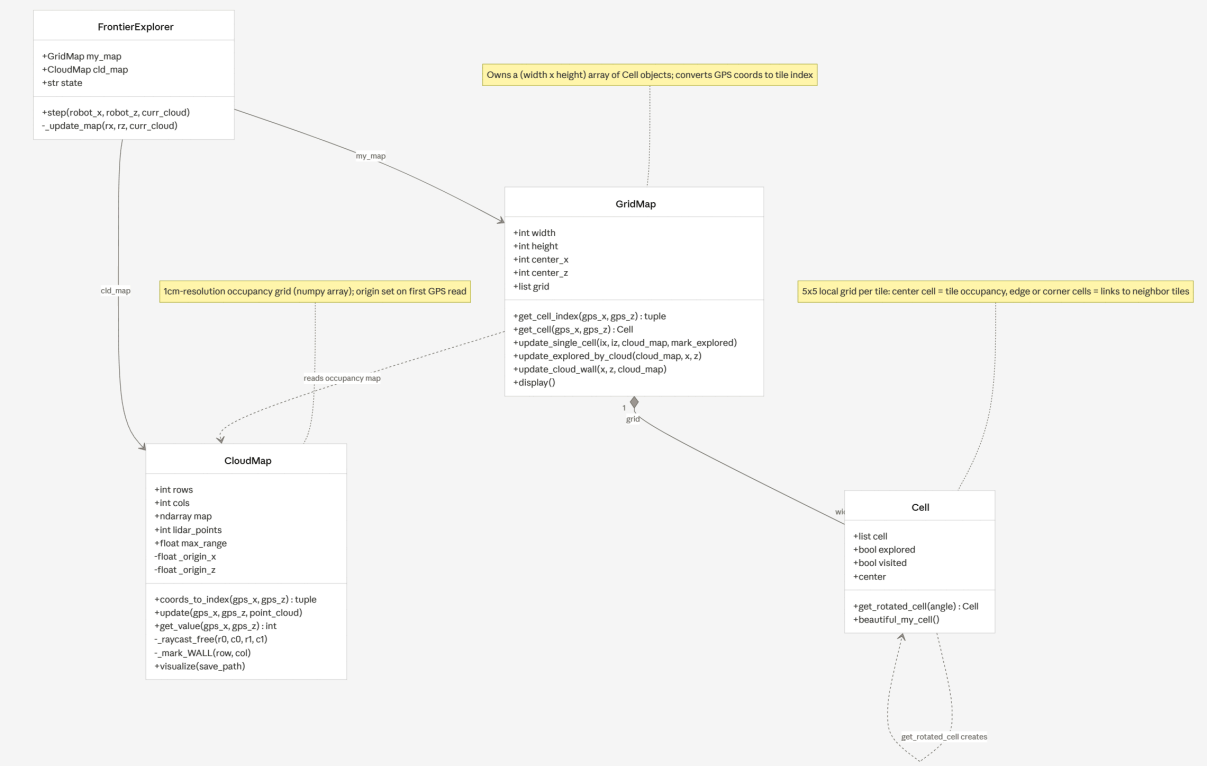
All other processes remained the same, with the exception of the preprocessing step. The original Histogram Equalization adjusts contrast based on global brightness information, which caused victim text to become degraded in environments with significant lighting variation. To address this, we switched to Adaptive Equalization, which adjusts contrast independently for each local region, resulting in a dataset better suited for training. Additionally, since the original augmentation settings applied rotation

and blur too aggressively — making the newly added low-quality images nearly illegible — we adjusted the settings to Rotation: Between -30° and +30° and Blur: Up to 6px. We also added 90° Rotate: Clockwise, Counter-Clockwise, and Upside Down to account for victims placed at 90-degree rotational intervals in the Webots map.

Training was conducted on Google Colab using YOLOv5, achieving an initial overall model accuracy of 0.871. However, when tested in the actual Webots environment on a custom map containing 15 victims placed at various angles, only 3 were successfully detected. When victims were placed at tilted angles, none of the three types were detected at all. Among the non-tilted victims, omega was detected but misclassified, psi was not detected at all, and phi was correctly identified approximately 50% of the time.
After applying the dataset improvements and retraining, the overall model accuracy slightly decreased to 0.865. However, in the actual Webots environment, the updated model successfully detected all 15 victims on the same map, correctly identifying both rotated and upright victims. This confirms that real-world performance is a more meaningful measure of reliability than benchmark accuracy alone, and that dataset quality and diversity have a greater impact on detection performance than raw accuracy scores.

6. Mapping + Implementation

6-1. Architecture design (with diagrams such as flowchart, UML, pseudocode)



UML diagram About Mapping System

The mapping system is built around three core classes — Cell, GridMap, and CloudMap — each responsible for a distinct layer of spatial representation.

A Cell is a class that stores all information pertaining to a single 12 cm × 12 cm tile. Internally, it is structured as a 5×5 grid and conforms to the mapping data transmission standard used for inter-module communication. Each Cell holds wall information, floor type information, and two state flags: `explored` and `visited`. The `explored` flag indicates whether more than 90% of the CloudMap points covering that Cell have been recorded, while `visited` indicates whether the robot has physically reached the center of that Cell.

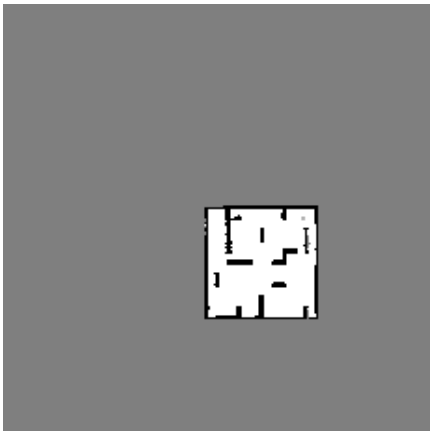
GridMap is a two-dimensional array map structure of size (2n−1) × (2m−1), composed of Cell objects. Given a pre-specified map size (n, m), the robot's starting position is set as the center of the grid, enabling a relative origin

system that allows dynamic adaptation regardless of the direction in which the robot explores.

CloudMap stores a pixel-level occupancy map at 1 cm × 1 cm resolution using LiDAR point cloud data. Each pixel is assigned one of three states: `_UNDEFINED`, `_WALL`, or `_FREE`. Like GridMap, it is initialized based on the input map dimensions (n, m), accounting for the 12 cm × 12 cm tile size, with the robot's starting position placed at the center. For each tick, one horizontal layer of LiDAR range data is processed across a full 360-degree sweep. Using the `skimage.draw` library, a ray is cast for each beam; the endpoint is marked as `_WALL`, while all points along the ray between the robot and the endpoint are marked as `_FREE`.

The CloudMap data is then used to populate the wall information in the GridMap. For each Cell, the corresponding region of the CloudMap is examined, and any pixel marked as `_WALL` is reflected into the Cell's internal 5×5 grid as a wall entry.

Floor tile type is determined by reading the color sensor. When a color tile is detected, the current zone is simultaneously updated and verified. Wall data and floor data are then consolidated — combining edge and corner information from adjacent Cells according to the mapping transmission standard — and the final map data is transmitted.



CloudMap output image

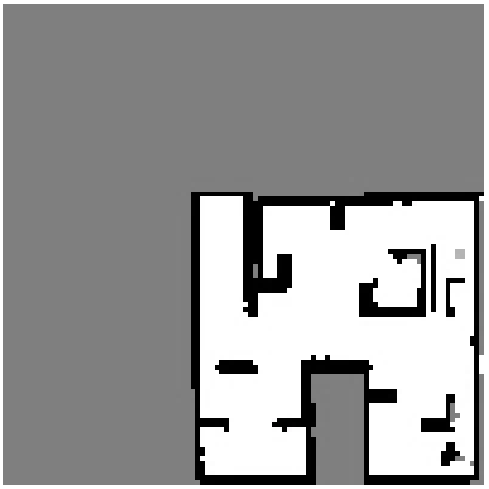
GridMap output image

6-2. Research and Analysis

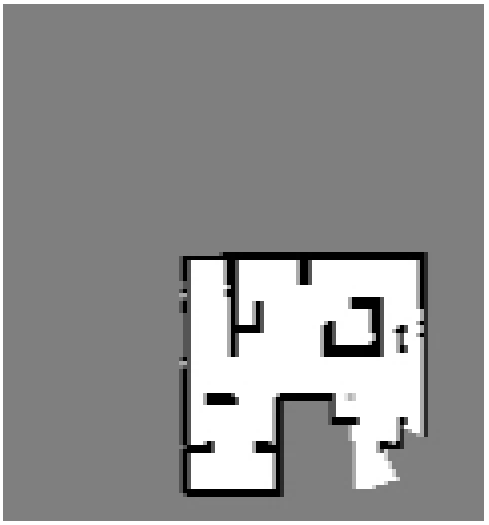
As described in the navigation section, the approach used at the February 2026 competition relied on tile-by-tile movement, with wall mapping performed via LiDAR point cloud data collected at each tile center. This meant that wall information was captured only at discrete, non-contiguous positions, producing a fragmented representation with significant gaps between sampling points. Since the revised navigation system required continuous, high-resolution wall data to support subgraph construction and fine-grained pathfinding, developing a continuously updated CloudMap became a foundational requirement.

The adopted approach marks wall information at 1 cm resolution in the CloudMap and subsequently transfers it into the corresponding GridMap Cells. To manage noise and prevent error accumulation from distant readings, the effective LiDAR range is capped at 25 cm from the robot center — slightly more than two tile widths. Any hit detected beyond this distance is not marked as `_WALL`, keeping the map clean and minimizing long-range inaccuracy.

An important finding was that updating the CloudMap on every tick, or whenever the robot entered a new tile, caused errors to accumulate and walls to appear thicker than their actual physical extent. To address this, CloudMap updates are now performed only upon arrival at a frontier, significantly reducing noise buildup.



CloudMap output before fix - thick-walled

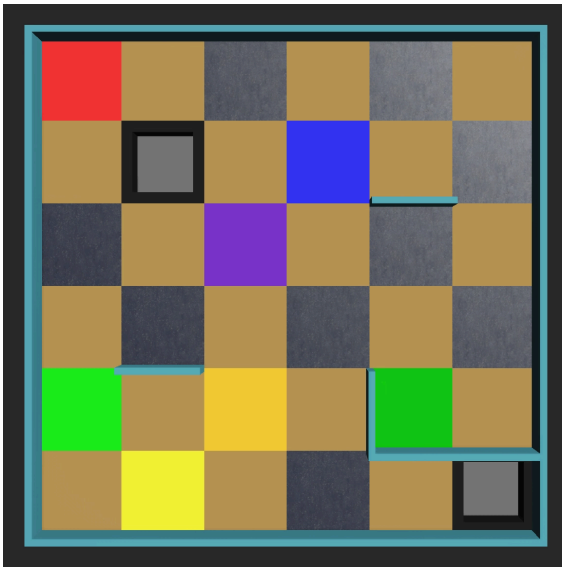


CloudMap output after fix - thin-walled

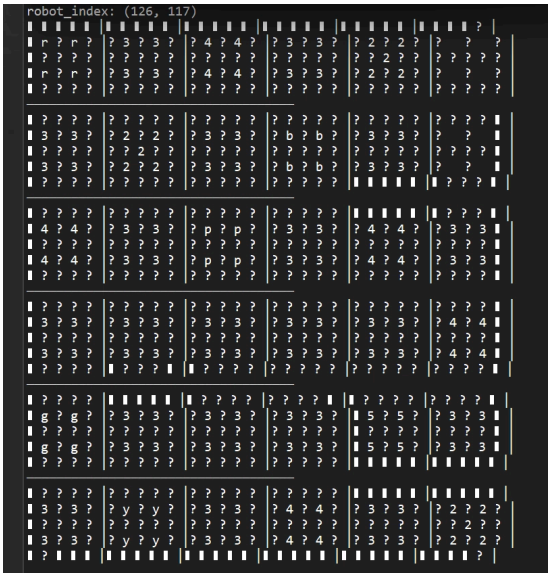
Color Tile Mapping Correction

A positioning discrepancy was identified in the color tile mapping process. Because the GPS receiver is located at the robot's center while the color sensor is mounted at the robot's front, the two can be observing different tiles simultaneously. This mismatch caused a single swamp tile, for example, to be recorded multiple times across different cells, effectively duplicating terrain features. To resolve this, the navigation module was updated to perform color tile mapping exclusively when the robot is at a tile center node, eliminating spurious multi-tile detections entirely.

On a color tile test map designed for accuracy verification, all tiles were mapped accurately with the exception of two.



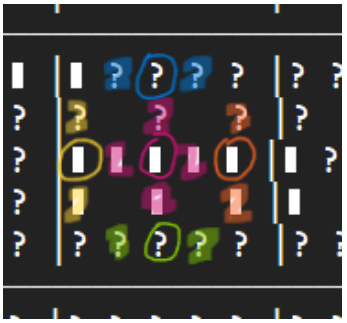
color tile test map world image



GridMap output of color tile test map

Wall Index Correction in GridMap

When transferring CloudMap data into the GridMap, it was observed that half-tiles and narrow passages were occasionally mapped inaccurately. Specifically, the 1st and 3rd index positions along each wall edge were prone to incorrect values due to the limited pixel coverage at those positions. A correction step was introduced in the `beautiful_my_cell()` method: for each of these intermediate wall indices, the values of the neighboring positions (adjacent cells or the center region) are checked, and the index value is adjusted accordingly to ensure structural consistency.

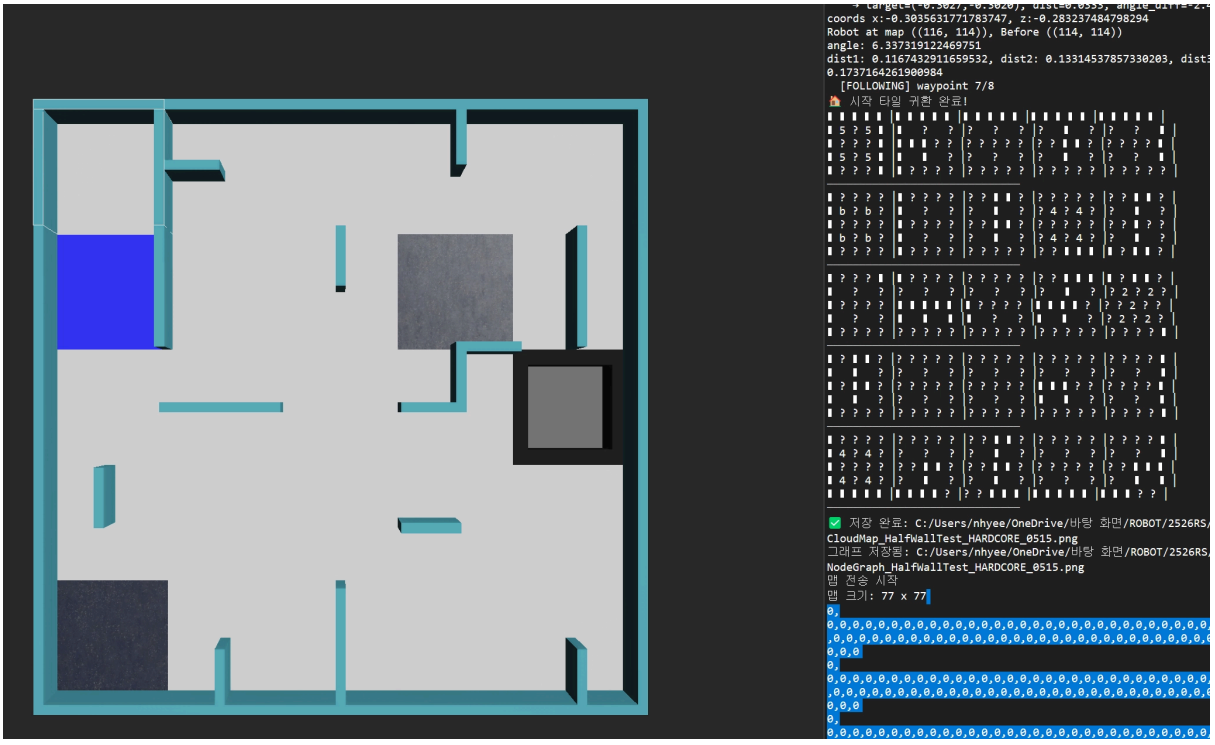


Explanatory image for the `beautiful_my_cell()` method

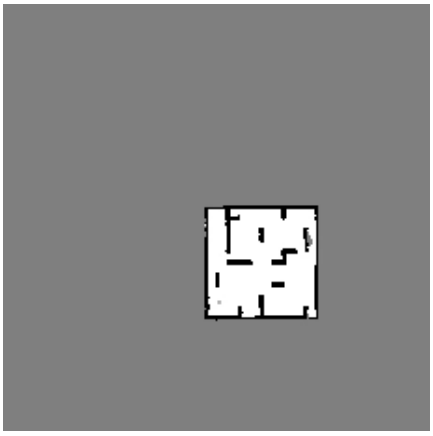
In this image, if the section marked with a circle is empty, the section sharing the same color must unconditionally be cleared.

6-3. Reliability Tests and quality assurance

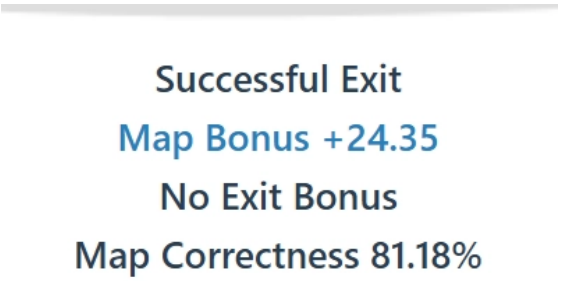
(Various mapping output images attached)



world map image & GridMap output

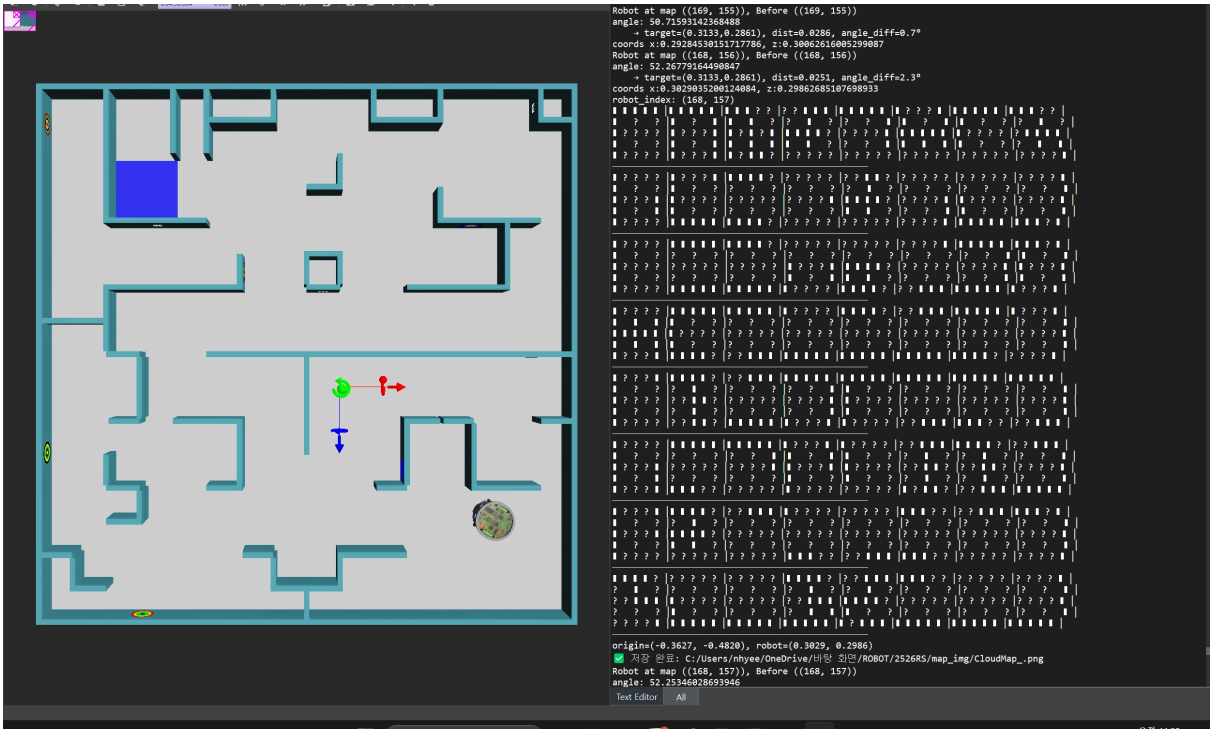


CloudMap output

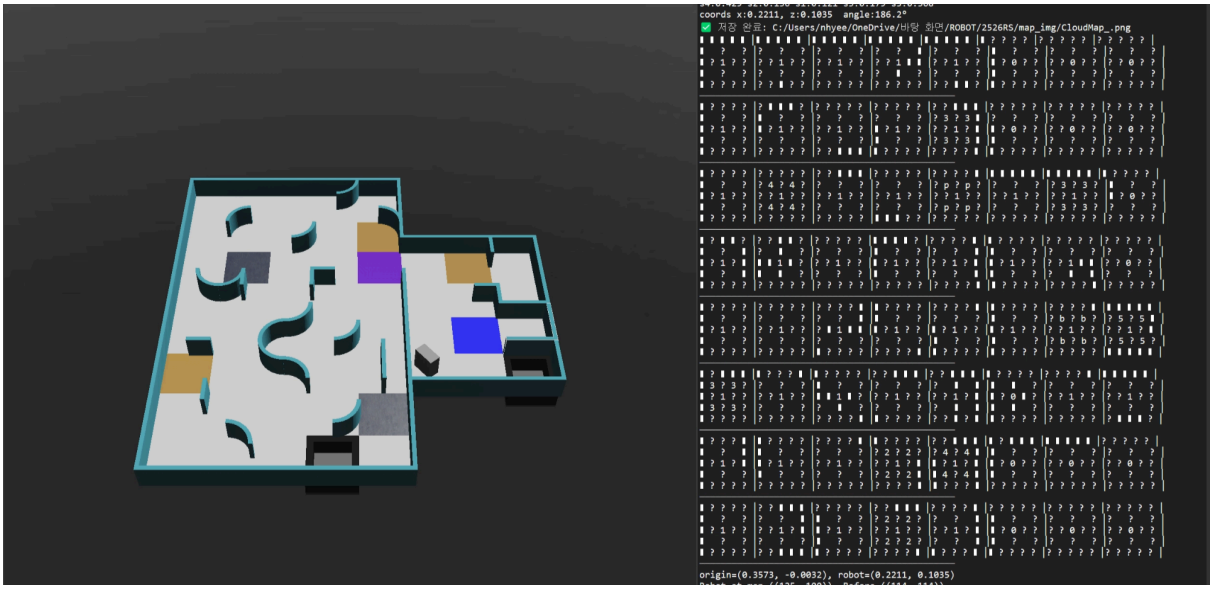


Erebus Simulation controls capture

Reliability testing focused primarily on positional consistency — verifying that the structural layout of walls as recorded by the robot matches the actual physical wall configuration of the test maze. Across multiple test runs on different map configurations, the system achieved an accuracy of approximately 80% when evaluated on a custom-built test map. Furthermore, the mapped wall structures were found to be in general agreement with the ground truth, confirming that the CloudMap-to-GridMap pipeline produces spatially accurate representations suitable for navigation.



world map image & GridMap output of only half-tile map



world map image & GridMap output of Curved Wall Test map

Other test runs also demonstrate that the environment is precisely and clearly mapped into the GridMap.